Care Delivery Facility-Class Design Basis

The First Facility Class of the Healthcare and Public Health Sector Design Basis

DBA-HPH-CD-FC1 — Facility-Class Design Basis, DBA-HPH Series

Issued Under: Healthcare and Public Health Sector Framework (DBA-HPH_Sector_Framework v0.5); DBA-MA Parent Framework

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Tim Roxey

Eclectic Technologies

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for care delivery, the first facility class in the DBA-HPH series. The *Healthcare and Public Health Sector Framework* (DBA-HPH_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility: the place where patients are cared for. It names the events a hospital must withstand, traces how each event reaches the patients inside it and the patients still on their way to it, sets the parameters that size the hospital's protection, and states what the hospital must be able to do on its own when outside power, water, communications and help are cut off. It describes no particular hospital. A design basis for a specific hospital, health system or state starts from this document and adds only what is particular to it.

The framework calls healthcare a hybrid sector and care delivery its consumer function: a hospital produces no commodity, and its resilience rests on what it inherits from other sectors (framework §1.5). It leaves to this class the calibration of the patient-care continuity window it proposes as the sector's consequence measure (framework §3.6), and the hospitals' side of the patient interface with the Emergency Services series, which it states at §6.4. This document takes up both. On the author's decisions of 12 October 2026 the class is written for hospitals as receiving facilities in full, with dialysis clusters and long-term care stated at the depth of the sub-index for a later revision to develop (Section 2.2).

1.2 Scope

The class covers the facilities in which patients are cared for, as the framework's register names them: hospitals and medical centers, dialysis clusters, and long-term care (framework §3.2, §8.2). For hospitals it includes the buildings and the spaces where care is given, from the emergency department to the operating rooms, the intensive care units and the wards; the essential electrical system and the generators that feed it; the water, heating and cooling plant; the piped medical gas and vacuum systems; the hospital's own communications; and the clinical and business information systems on which care now runs, from the electronic health record to the laboratory, imaging and pharmacy systems. It includes the hospital's capacity to receive patients: its emergency department, its surgical capacity and the beds behind them.

Three neighbors are separate classes of this series. The production and supply of medical materials, from pharmaceutical manufacture to the blood supply and the stockpiles, is *Medical Materials and Supply* (DBA-HPH-MM-FC2). Public-health laboratories, disease surveillance and mass-care capability are *Public Health and Laboratory* (DBA-HPH-PH-FC3). The coordination of the sector, through the federal preparedness structure, the Health-ISAC and the health information exchanges, is the coordination class, DBA-HPH-OpsCo. They are covered here only where the hospital depends on them. Where a hospital runs its own laboratory, the laboratory is this class's; where it relies on a shared laboratory service outside its walls, the loss

of that service is a design-basis event here (Section 5.7) and the service itself is the class that produces it.

Two boundaries with other series are drawn in the framework and are taken here as given (framework §6.4). The pre-hospital response belongs to the Emergency Services series and the receiving facility to this one, and the point at which a patient passes from a crew to the hospital is the interface both series state. Medical control is divided on the same line: the clinical authority of the medical director is this sector's, and the dispatch and field actions taken under it are the Emergency Services series'. The substrate the hospital stands on, electric power, water, communications, is governed by the series that produce it, and the hospital inherits its physical design basis from them (framework §8.1). What this class adds is the hospital's own provision against the loss of that substrate, and the consequence of the loss to the patients inside. Section 8.3 states the boundaries.

What makes a facility a member of this class is the same at every site: it holds people whose lives depend on care that must continue, and many of them cannot leave on their own. The particular building, population and services differ from hospital to hospital, and that detail belongs in each hospital's own design basis.

1.3 The Reference Facilities

Three kinds of hospital are used as reference cases. Each was chosen because it shows a different way the care a hospital gives can be lost or overwhelmed.

The Level I trauma center in a metropolitan area is the hospital the area's plans send its most seriously injured to. It carries the surgical capacity, the blood supply and the specialist staff that definitive care for trauma requires, and it is the governing configuration for the surge chain, because its throughput to surgery sets how many of the critically injured are treated in the time that matters. The trauma verification standards of the American College of Surgeons require such a center to have an operating room staffed and available within fifteen minutes of notification, and a second when the first is occupied, but set no rate at which patients are to pass through (Section 3.3). It is also the hospital whose loss in a conflict would remove capacity no other hospital in the area can replace, which is why the framework's presumption toward military-action applicability is strongest for trauma and burn capacity (framework §7).

The community hospital nearest a likely scene is a general hospital without trauma designation, standing closest to a place where people gather. It bounds the receiving path. In a mass-casualty event the injured go to the nearest door, by any means they can find, and the plans of the response do not decide where they arrive. In Las Vegas in October 2017 the nearest hospital, which is not a trauma center, received about 150 patients (framework §6.4; *Route 91 Harvest Shooting* F-4). The community hospital must therefore receive a surge it was not designated for, without notice and without field triage, and stabilize and pass on the patients it cannot treat.

The multi-hospital system on one shared electronic record is a group of hospitals, often across a region or several states, whose clinical and business systems are run as one. It bounds the

information path. When the shared systems are lost, every hospital in the system loses them together, and each falls back to paper at the same moment. The ransomware attack on the Irish Health Service Executive in May 2021, which took the information systems of a national health service out of use for months, is its case (Section 5.5).

A requirement that names one of the three applies to that kind of hospital. A requirement that names none applies to all three. Any other hospital follows a requirement to the extent it shares the services, setting or systems the requirement addresses. Dialysis clusters and long-term care facilities follow a requirement where Section 2.2 says they do.

1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (*Facility-Class Types Are Physical, Coordination and Logical*). The harm it governs is the loss of care at a place where patients are, and that harm runs through the building, its plant and the people inside it, even where the cause is the loss of an information system. In the framework's structure it sits under Care Delivery (CD), the first of the sector's three function classes (framework §3.1, §3.2).

1.5 Relationship to Governing Documents

This instrument is issued under the Healthcare and Public Health Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, consequence measure or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form "framework §N" resolve against the Healthcare and Public Health Sector Framework in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term "envelope". Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Care Delivery Facility Class

2.1 Defining Characteristics

For design-basis purposes, the hospital is defined by three features. First, it holds people who cannot leave. A patient on a ventilator, on an operating table or in a neonatal incubator depends on care that must continue without interruption, and many patients on the wards could not leave the building without help even if it were safe to try. When the care stops, the harm falls on them where they lie, and moving them is itself a hazard. Second, it receives. A hospital cannot choose its patients the way a factory chooses its orders. Federal law requires a hospital with an emergency department to give an appropriate medical screening examination to anyone who comes to it (42 CFR 489.24(a)), and in a disaster or an attack patients arrive by their own means, without notice, in numbers no one has planned for. Third, it consumes. Its power, its water, its medical gas, its communications and now its information systems come from outside, and each is converted inside the building into care that must not stop. The loss of any one of them is measured in hours, not days, for the patients who need it most.

People who cannot leave, a door that cannot close, and a dependence on inherited supplies measured in hours: these three define the class. They set it apart from the medical-materials class, whose plants produce goods that can be stockpiled, and from the public-health class, whose harm is the loss of a population's detection and response rather than of a patient's care.

2.2 The Time-Criticality Sub-Index

The class is sub-indexed by life-safety time-criticality, as the framework requires (framework §3.1, §8.2), so that the design-basis load a facility carries scales with how quickly its failure harms the patients who depend on it. Three bands are used. These bands refine the framework's sub-index, which places dialysis and critical care together in hours and names dialysis the most time-critical (framework §3.2, §3.6); the refinement separates care harmed within minutes of the loss of power or gas from dialysis, whose facility window is set in hours by its water and power but whose patients are harmed over a longer interval, and it is noted to the framework (Section 10.10). The first, life-sustaining care, is care whose interruption harms a patient within minutes: ventilation, anesthesia and surgery in progress, intensive and neonatal care. The second, time-critical care, is care whose interruption harms a patient within hours: dialysis, the emergency department's definitive care for the injured, and the surgery that follows it. The third, general care, degrades over a longer interval but still fails, through the loss of medication, nutrition, temperature control and sanitation. Every hospital carries patients in all three bands, and the continuity window of Section 3.3 is stated for each band it carries. The framework calibrates the window to the most time-critical population a facility serves (framework §3.6); a hospital states its window for that population first and for the others beside it.

Dialysis clusters are stated here at the depth of the sub-index, on the author's decision of 12 October 2026. A dialysis facility's governing dependency is dialysis-grade water, the

dependency the water framework names as the most time-critical in its sector (Water Sector Framework §8.2). Federal conditions for dialysis facilities incorporate an industry standard of 2004 for water and dialysate quality (42 CFR 494.40, incorporating ANSI/AAMI RD52:2004) and require the facility's emergency plan to address the interruption of the water supply and the failure of equipment and power, and to train patients to disconnect themselves from the machine (42 CFR 494.62). The facility's continuity window is set in hours by the endurance of its water and power against its treatment schedule, and its patients' by the interval they can safely go without treatment, which is longer; its design basis is the first set against the second. Its development is for a later revision of this class (Section 10.3).

Long-term care is stated at the same depth. A nursing home or rehabilitation center holds residents who are frail and often cannot leave without help, in a building with less clinical support than a hospital. Federal conditions for long-term care already require emergency power and policies for maintaining temperatures (42 CFR 483.73(b)(1)(ii)), but the provision is smaller than a hospital's and the residents are as vulnerable to its loss. Its governing hazard is the loss of temperature control, which can harm residents within hours in extreme heat or cold, as the deaths at a Florida rehabilitation center after Hurricane Irma in September 2017 showed, where the building kept power but lost its cooling, as reported. Its development is likewise for a later revision (Section 10.3).

2.3 Deployment Typologies and Size Range

The class spans hospitals owned by systems, governments, universities and religious and community bodies, from the regional medical center with several hundred beds to the critical access hospital with a few. Many now belong to systems that run their information services as one, so that a single failure reaches every hospital in the system together, which is why the third reference facility is a system and not a building. The class applies across the size range. A small hospital nearest a scene can do more harm by failing to receive than a large one far away, and what counts in this framework is the severity of the harm, not the size of the facility (framework §4.1).

Hospitals also form regional clusters whose surge capacities are counted together. When one hospital loses its power or goes on diversion, its patients go to its neighbors, which must absorb them on top of their own. In the New York evacuations during Hurricane Sandy in 2012, the city's hospitals absorbed the patients of several evacuated hospitals at once (Adalja and colleagues, *Absorbing Citywide Patient Surge During Hurricane Sandy*). The cluster raises the question the refinery and dam classes ask of their own clusters: what happens if several are lost together. For hospitals the answer is that a regional event takes the neighbors' capacity at the same moment it creates the demand for it.

2.4 Consequence Context

The hospital has two governing consequences, and they run in opposite directions. The first is the loss of care to the patients inside. When the power, water, medical gas or information

systems on which care depends fail beyond the hospital's own provision, the care stops, and the patients in the most time-critical band are harmed first. The framework measures this as a duration, the patient-care continuity window (framework §3.6). The second is the failure to receive. When more patients arrive than the hospital can carry to definitive care in the time that matters, the patients beyond its capacity wait, and some of them die waiting. This consequence is measured as a rate, not a duration: how many critically injured patients the hospital can carry through to definitive care in an hour (Section 3.3).

A regional disaster or an attack can open both at once. The event that takes a region's power and communications can also produce its casualties, and the hospital then receives a surge with its own care already degraded and its notification cut. That compound is the governing military-action scenario of Section 6.3, and it is the scenario that DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*) traces through the class (Section 9.2).

3. The Facility-Class Design Envelope

3.1 Why Care Delivery Is the First Class

The framework orders care delivery first in its register and names the hospital and the dialysis cluster as the facilities whose dependency the water and electric series already elevate (framework §1.5, §6.3). This document takes that ordering as given: the hospital is the facility other series' design bases already protect, and this class is the supply-side dual that states what the hospital itself must do. The range of events the hospital must withstand is set by the two ways its failure reaches patients, described in Section 3.2.

3.2 The Two Consequence Chains

Chain C — continuity. The loss of electric power, water, medical gas, heating or cooling, communications or the information systems on which care runs, beyond the hospital's own provision, interrupts care to the patients inside. The chain runs through the patients in order of their time-criticality, on the timescale of the continuity window of each band. It ends, if the provision cannot be restored, in the evacuation of the hospital, which is itself a hazard to the patients least able to be moved. It is the governing chain for the inside of the hospital.

Chain S — surge. More patients arrive than the hospital can carry to definitive care in the time that matters to them. The chain runs from the scene, through whatever route the patients take, to the emergency department, the operating rooms and the beds behind them, on the timescale of minutes to hours. It is the governing chain for the hospital's receiving function, and its consequence is set by the throughput of Section 3.3, not by the count of beds. It also runs between hospitals: a hospital that goes on diversion or evacuates moves its demand to its neighbors and opens their Chain S.

The two chains meet at the hospital. A hospital that loses its substrate cannot receive at its rated throughput, and a hospital that receives a surge consumes its power, water, gas, supplies and

staff faster than its continuity provision was sized for. The design basis carries both chains, and Section 6.3 is the scenario in which both open together.

3.3 Design Parameters at Class Resolution

Three parameters govern the envelope, on the author's decision of 12 October 2026. The first is the continuity window: for each band of the sub-index the hospital carries, the interval within which its care must be sustained without the substrate the event removes, before patients in that band are harmed (framework §3.6). It is stated against each substrate separately, because each fails differently and each has its own provision: electric power against the endurance of the essential electrical system and its fuel; water against on-site storage and the hospital's ability to operate without municipal pressure; medical gas against bulk storage, reserve supply and the time to resupply; and the information systems against the hospital's manual mode, which Section 5.4 requires to be measured. The window is the shortest of these for the band, because the patient is harmed by whichever fails first.

The second is the surge throughput: the number of critically injured patients the hospital can carry from arrival through triage and the emergency department to definitive care, surgery for most of them, in an hour, sustained across the surge, limited by whichever of imaging, blood release, anesthesia, surgeons or operating rooms is exhausted first. The framework records from the Las Vegas surge that the hospitals' ability to move patients quickly through triage and the emergency department to surgery mattered more than their count of surge beds (framework §6.4; *Route 91 Harvest Shooting* F-4), and the throughput is the measure that captures it. No federal or professional standard sets a rate. The trauma verification standards set the time within which an operating room must be staffed and a second room available (American College of Surgeons, *Resources for Optimal Care of the Injured Patient*, 2022 standards, Standards 3.1 and 3.2), which bounds the first patient's wait but not the rate. The peer-reviewed literature offers a measure of acute-care surge capacity as the emergency department's beds divided by the time each patient spends in them, expressed in critical and moderate casualties per hour (Bayram, Zuabi and Subbarao, *Disaster Medicine and Public Health Preparedness*, 2011), which the class takes as a starting point and not as a standard. The throughput is stated for the hospital as it would actually be staffed at the hour the surge arrives, and for the surge that arrives without field triage. The throughput is stated on generator power. Every load the throughput needs, the imaging, the blood bank's refrigeration, the sterilizers, and the operating rooms' lighting and ventilation among them, is named together with the branch of the essential electrical system that carries it, and the throughput is the one the system can sustain with those loads added to the load the event already imposes. A load the throughput needs that is not on a branch the system carries at that load is counted as lost, and the throughput is stated without it. The endurance of the essential electrical system at that load is stated under Section 5.1.

The third is the time to surge posture: the time from the first sign of a surge, whether a notification or the first patient at the door, to the point at which the hospital is receiving at its stated throughput, with its incident command established, its operating rooms cleared and

staffed, its triage set at the entrance and its staff recalled, under the rule of Section 4.6 where no message can reach them. It is stated for a surge that arrives without notice, because the receiving hospital may learn of the event from its first patient. The federal Hospital Preparedness Program's exercise presents a coalition with a surge equal to twenty percent of its staffed beds and sets no time limit (Assistant Secretary for Preparedness and Response, *Medical Response and Surge Exercise*), and one state's guidance sets twenty percent within four hours of notification (Michigan Department of Health and Human Services, *Immediate Bed Availability* guidelines, 2018); neither counts the time to throughput, which is what the class measures.

This instrument states the parameters and their governing logic. Their quantitative values await validation against the plans and exercises hospitals already hold: their emergency operations plans, their hazard vulnerability analyses, their exercise reports and their trauma activation records (Section 10.1). They are not asserted here as numbers.

4. The Receiving Function

4.1 Status and Use

The framework states the hospitals' side of the patient interface with the Emergency Services series, on the author's decision of 7 October 2026, and leaves its development to the first receiving-facility class (framework §6.4). This section develops it. It is the hospital's half of a two-way interface whose other half is stated in *Response Forces and Stations* §4.7 and §8.3, and the two are read together. On the author's decision of 12 October 2026 the Las Vegas hospital surge is cited here from the Emergency Services series, where it is conferred as DBA-ESS-RE-11; it keeps its single designator, and healthcare is to be entered as an impacted sector in the corpus register, which is Project 0's (Section 10.4).

4.2 Patients Who Arrive by Their Own Means

The receiving hospital plans for a surge that arrives without notice and without field triage. In Las Vegas, on the Nevada Hospital Association's case study as the framework records it, about eight hundred of the injured found their own way to hospitals, carload after carload arriving without notice, and five hospitals received about 750 patients between them (framework §6.4). A plan that assumes patients will be triaged in the field and distributed by the response is a plan for a different event. The class therefore requires triage at the hospital's entrance, set up from the first sign of a surge and staffed by people who can sort patients arriving in private vehicles as well as by ambulance; a route through the hospital from that entrance to the operating rooms that does not pass through registration first; and registration under temporary identities that keeps pace with arrivals rather than holding them. In Las Vegas patients were treated under aliases faster than registration could keep up (framework §6.4).

The duty to screen is not suspended by a surge. A hospital may ask the response to send ambulances elsewhere while it is in diversionary status, but a patient who comes to its emergency department by any other means must still be screened (42 CFR 489.24(a), (b)(4)).

Where the Secretary has invoked a section 1135 waiver for the area, sanctions for an inappropriate transfer or for screening at an alternate site do not apply, but the waiver runs for seventy-two hours from the activation of the hospital's disaster protocol, except in a pandemic (42 CFR 489.24(a)(2)). The class treats the self-presenting patient as part of the design-basis surge and not as an exception to it.

4.3 The Count Built from the Receiving End

The receiving hospital counts and reports the patients who reach it, by whatever means they came (framework §6.4). Command at the scene requires the count of patients by hospital, and since most of the patients in a surge may never pass through the response, the count can only be built from the hospitals (*Response Forces and Stations* §4.7). The class requires each hospital to report its arrivals, their acuity and its remaining throughput to the area's command at a stated interval from the first sign of a surge, by a path that does not depend on the hospital's information systems being available, and to keep reporting until the surge has passed. The count is what lets the response send the next patients to the hospital that can still take them, and it is the hospital's contribution to the distribution of patients the two series share. The class requires each hospital to name the path that remains when the public networks are gone, and to state whether it is a terminal on the area's public-safety radio system or a path of the hospital's own that does not depend on the area's ground networks, such as a satellite terminal or a station of the area's amateur radio emergency service. Where it is a terminal on the area's public-safety radio, the hospital is a user of a system that *Public-Safety Answering and Dispatch* governs, and the terminal, its power from the essential electrical system and the people trained to work it are part of the hospital's design basis; the joint between the two series is noted to the Emergency Services series (Section 10.10). The path is credited only once it has been used in an exercise with the public networks out of service.

4.4 Notification That Does Not Depend on an Answer

The receiving hospital is able to receive notification from the response by a means that does not depend on someone answering a radio (framework §6.4). In Las Vegas several hospitals did not answer the radio when the fire department's dispatch center first tried to notify them (framework §6.4; *Route 91 Harvest Shooting* F-4). The class requires a notification path that alerts the hospital's emergency department and its incident command at once and records that it has been received, and a second path that does not share the first's network or its point of failure. Where the event has taken the area's communications, the hospital's own first patient is its notification, and the time to surge posture of Section 3.3 is counted from that arrival. Where the area's command can still reach the hospital once the public networks are gone, it reaches it by the path named in Section 4.3, and that path is the second path here unless the hospital holds another that meets the same test.

4.5 Diversion and Transfer as Actions

A hospital that goes on diversion, or that transfers patients out to make room, moves its demand onto its neighbors. Diversion is decided by the hospital under the region's emergency medical services rules, and federal law permits it for ambulances not yet on hospital property (42 CFR 489.24(b)(4)). The class treats a decision to divert, or to transfer patients out during a surge, as an action placed on its consequence: the time it adds to the journey of the patients it redirects, and the load it places on the hospitals that receive them. Where a hospital's decision to divert is supported or made by a system within the scope of *Layer Decomposition and Determinism* §4.3, the placement of Section 7.3 applies to it. A diversion taken because the hospital has lost its information systems, as several hospitals of a system did during the Ascension ransomware attack of May 2024, as reported, is a Chain C event whose consequence reaches the neighbors through Chain S.

The receiving function also ends somewhere. A community hospital that receives patients it cannot treat stabilizes and transfers them, and its transfer path, the ambulances and the receiving trauma center, is part of its throughput. The class requires a hospital that is not a trauma center to state the rate at which it can stabilize and pass on the critically injured, and the path by which it does so when the area's ambulances are committed to the scene.

4.6 Staff Recall When No Message Can Be Sent

The time to surge posture includes the recall of staff (Section 3.3), and the governing scenario requires that recall not depend on the networks the campaign denies (Section 6.3). When the paging system, the telephones and the messaging services are gone, no message can be sent, and the only recall that works is one that needs none. The class therefore requires a standing rule of self-activation. While the hospital's applicability determination is positive and the public networks are down, staff who learn of a mass-casualty event in the area, by broadcast radio, by word of mouth or from the event itself, report to their hospital without being called. The rule is written into the hospital's emergency operations plan, taught to the staff it binds and exercised, because a rule the staff do not know does not recall them. The time to surge posture is stated with the rule in force and counted from the first patient (Section 4.4).

The rule does not call everyone. A hospital that brings in all its staff at once has no one to relieve them, and the class names its people as the limit of its endurance (Section 6.8). The rule therefore divides the staff in advance, by role, into those who report at once and a designated reserve, who stay away, rest, and report at a stated later time to relieve the first, whatever they learn in between. Those who report at once go to a labor pool at a stated place in the hospital, where its incident command assigns them, rather than to their usual units, so that the surge takes the staff it needs and the units keep theirs. The division is set so that the relief arrives before the hold time of the manual mode runs out (Section 5.4). A hospital whose applicability determination is negative is not bound by the rule, but one whose hazard vulnerability analysis includes an event

that takes the public networks and produces casualties together, an earthquake for instance, faces the same problem and should adopt it.

5. Design Basis Events: Non-Adversarial Envelope

The framework lists candidate bounding events for the sector but does not yet develop them (framework §4.1). This section develops each for the hospital, using the series' four headings: the event, the harm that makes it count, what the hospital must do, and for how long. Durations are stated as class parameters awaiting the framework's values (Section 10.2).

5.1 Loss of Electric Power Beyond On-Site Endurance

Postulated event loss of grid power for longer than the hospital's essential electrical system can carry its load, whether because the outage outlasts the fuel, because the generators or their fuel pumps, transfer switches or switchgear are lost to the same event, or because fuel cannot be delivered.

Consequence threshold loss of power to life-sustaining care, the operating rooms or the systems that keep the building habitable, within the continuity window of the most time-critical band (Chain C).

Required function the essential electrical system must carry its load for the continuity window the hospital states, with every component on which it depends, the generators, their fuel, fuel pumps and day tanks, the transfer switches and the switchgear, located and protected so that the event that takes the grid does not also take them. Federal conditions require a generator installed in new construction or renovation to be located, and the system inspected, tested and maintained, in accordance with the fire and electrical codes the conditions adopt, and a hospital with an on-site fuel source to have a plan for keeping its emergency power operating during the emergency unless it evacuates (42 CFR 482.15(e)). No federal rule sets the quantity of fuel. The Joint Commission requires a plan for how the hospital will manage for ninety-six hours, and states that the hospital is not required to remain fully functional for that period or to stockpile supplies for it (Joint Commission, *Emergency Management* standards, effective 1 July 2022, EM.12.02.09, EP 3). The class requires the hospital to state the endurance of its essential electrical system as a number of hours at the load it would carry in the event and again at that load with the surge loads of Section 3.3 added, the fuel it holds, the delivery on which any longer endurance depends, and whether that delivery survives the event.

Design-basis duration the continuity window of the most time-critical band the hospital carries; for the endurance of on-site provision, the longest outage the design-basis event can produce, which the hospital's hazard vulnerability analysis informs but does not cap, until power is restored or the patients are moved.

This category is characterized empirically by the loss of power at Memorial Medical Center in New Orleans during Hurricane Katrina, conferred by the framework as DBA-HPH-RE-2, an

event distinct from the failure of coordination in the same storm that the Emergency Services series holds as DBA-ESS-RE-8 (framework §9.1). On ProPublica's account, city power failed early on 29 August 2005 and the hospital's generators took over; floodwater rose the next day; and at about two in the morning of 31 August the generators stopped, the main transfer switches having been installed only a few feet above ground level, below the flood the storm brought. Without power or running water the temperature inside rose above a hundred degrees. About two thousand people had been inside at landfall, more than two hundred of them patients. The last living patient left on the evening of 1 September, and forty-five bodies were later recovered from the hospital. The design-basis reading is that the generators were not the weak point; the components between the generators and the load were, and they sat below the flood that the event brought. Seven years later, during Hurricane Sandy, the generators at Bellevue Hospital in New York were on upper floors but the pumps that fed them fuel were in the flooded basement, and National Guard troops carried fuel up thirteen floors by hand before the hospital evacuated its patients, as reported. The lesson had been learned about the generator and not about what the generator needs.

5.2 Loss of Water Supply or Pressure

Postulated event loss of the municipal water supply or its pressure, or contamination that makes it unusable, for longer than the hospital's on-site provision.

Consequence threshold loss of dialysis, sterilization, sanitation, fire protection, and of heating or cooling where the hospital's plant runs on municipal water (Chain C).

Required function the hospital must state every function that depends on municipal water and the endurance of its provision for each, including the functions that depend on water indirectly: boilers, chillers and cooling towers, sterilizers, and the hospital's own dialysis. A plant that cannot run without municipal pressure is a single point of failure for every function it serves. On-site storage, a second connection to a different part of the water system, and the means to receive water by truck are each credited only where they survive the event that takes the supply. Dialysis-grade water is stated separately, because its continuity window is the shortest in the water series (Water Sector Framework §8.2).

Design-basis duration the continuity window of the band each function serves; until supply is restored or the patients who depend on it are moved.

During the winter storm that struck Texas in February 2021, St. David's South Austin Medical Center lost its city water pressure on 17 February; its boilers depended on that water, so it lost its heat as well, and it moved some of its patients out, as reported at the time. The design-basis reading is that a hospital can lose its habitability through a utility it does not think of as life-sustaining, and that the dependency runs through the plant room, not only through the clinical areas.

5.3 Loss of Medical Gas or Its Supply

Postulated event failure of the piped oxygen, medical air or vacuum system, loss of the bulk supply, or interruption of resupply beyond the hospital's reserve.

Consequence threshold loss of oxygen or vacuum to patients in life-sustaining care, the operating rooms and the emergency department (Chain C).

Required function the medical gas and vacuum systems serving life-sustaining care must be protected as the essential electrical system is, with their alarms independent of the hospital's information systems, a reserve stated in hours at the load the event would bring, and a resupply path that does not depend on the roads, power or communications the event may take. A surge raises the consumption of oxygen in the emergency department and the operating rooms at the moment resupply is hardest, and the reserve is stated for the surge as well as for the normal load. The systems themselves are governed by the health-care facilities code the federal conditions adopt (42 CFR 482.15(h), incorporating NFPA 99, 2012 edition), and this class adds the reserve and the resupply.

Design-basis duration the continuity window of life-sustaining care; until resupply is assured.

5.4 Loss of the Hospital's Information Systems

Postulated event loss of the electronic health record and the systems that depend on it, the laboratory, imaging, pharmacy, ordering and the hospital's telephones where they run on the same network, through failure, a faulty update or a supplier's outage.

Consequence threshold care that cannot be ordered, checked or recorded, results that cannot be returned, and medication that cannot be safely dispensed, slowing every patient's care and halting what cannot be done by hand (Chain C), and a hospital that diverts ambulances because it cannot receive them safely (Chain S).

Required function the hospital must hold a manual mode for every clinical function that depends on its information systems and must state, for each, the rate at which it can be carried out by hand and how long that rate can be held before staff, paper and accuracy run out. This is the manual fallback rate and hold time the transportation series requires of its nodes (*Nodes and Terminals* §3.4), applied to care. The manual mode is exercised, and its rate is measured in exercises and in real events; a rate that has never been measured is not known. The hospital also holds, outside the systems that may be lost, what it needs to start the manual mode: the current list of patients with their medications and allergies, the downtime forms, and a way to return laboratory and imaging results to the clinician who ordered them.

Design-basis duration until the systems are restored and their records reconciled; for a failure, hours to days; for an attack, the duration of Section 5.5.

The faulty update to a widely used security product in July 2024 took Windows systems out of service around the world on a single morning, and hospitals cancelled non-urgent surgery for

about a day, as reported. It was not an attack, and no one resisted the recovery. It shows the shape of the event without its duration: every hospital that ran the product lost its systems at the same moment, and a hospital's recovery that morning depended on how ready its manual mode was.

5.5 Cyber Attack on Clinical and Hospital Systems

Postulated event an intrusion that encrypts, destroys or alters the hospital's information systems, or reaches the systems that control its plant or its medical devices, whether aimed at the hospital or at a shared system it depends on.

Consequence threshold loss of the information systems for weeks rather than hours (Chain C); diversion and the transfer of demand to neighbors (Chain S); and, where records, orders or device settings are altered rather than lost, care given on false information.

Required function the hospital must hold the manual mode of Section 5.4 for the duration an attack can impose, which is measured in weeks and months, not days. The plant controls, the essential electrical system's controls and the medical gas alarms must not be reachable from the networks on which the clinical and business systems run, so that an attack on the record does not also take the building. Medical devices whose settings can be changed over a network are governed under Section 7.3, each change of setting placed on its own consequence. Where the hospital belongs to a system, the design basis states which of its functions are lost when the system's shared services are lost, and the system states how each hospital will operate alone. The integrity of records on which a clinical decision rests is governed by *Indication Integrity*, and the class requires that a hospital restoring its systems after an attack be able to establish which records it can trust before it relies on them.

Design-basis duration indefinite; the condition persists until the systems are restored from a source the attacker could not reach and their records reconciled.

This category is characterized empirically by the ransomware attack on the Irish Health Service Executive in May 2021, conferred by the framework as DBA-HPH-RE-1 (framework §9.1). On the independent post-incident review commissioned by the Executive and published in December 2021, the attacker gained access through a malicious spreadsheet attached to a phishing email sent on 16 March 2021 and opened on 18 March and moved through the network for about eight weeks without being detected, and in the early hours of 14 May detonated ransomware that encrypted about eighty percent of the Executive's information technology environment. The Executive shut down its systems and disconnected its national network from the internet. Hospitals went to paper; on the first day, thirty-one of the country's fifty-four acute hospitals cancelled some services, on the review's account. The decryption key was obtained on 21 May, and the review records that about ninety-nine percent of applications had been restored by 21 September, more than four months after the attack. The review found no empirical data isolating the harm to patients. The design-basis reading is that the duration of the manual mode

was set by the time to restore a national estate, not by the time to decrypt it, and that a health service built its continuity for hours and was asked for months.

The attack on Ascension in May 2024 shows the same event in a system of hospitals across many states: its electronic health record and the systems for ordering tests, procedures and medications were lost, staff charted on paper for about five weeks, and ambulances were diverted from several of its hospitals, as reported. The ransomware attack on the University Hospital of Düsseldorf in September 2020 is often cited as the first death caused by such an attack; a patient redirected to a more distant hospital died, but the prosecutors concluded that she would likely have died in any event, and the class does not cite it as a death caused by the attack.

5.6 Mass-Casualty Surge Beyond Capacity

Postulated event an incident, accidental or deliberate, that sends more critically injured patients to the hospital, or to the hospitals of an area, than they can carry to definitive care in the time that matters, including a surge that arrives without notice and without field triage.

Consequence threshold critically injured patients who wait beyond the time that matters to them, and a hospital whose own care to its existing patients is degraded by the surge (Chain S, with Chain C touched).

Required function the hospital must receive at the surge throughput and reach surge posture in the time it states under Section 3.3, for a surge that arrives by every means, and must carry out the receiving function of Section 4. Its supplies for the surge, blood, airway equipment, linens and the rest, are stated for the surge's first hours without resupply; in Las Vegas linens and airway supplies ran short within hours (framework §6.4). Families arriving with patients are part of the load the hospital plans for; in Las Vegas four to six family members arrived for every patient (framework §6.4). The hospital's incident command is established under a recognized hospital incident command system, of which the *Hospital Incident Command System Guidebook*, fifth edition, published by the California Emergency Medical Services Authority in 2014, is the standard reference.

Design-basis duration the duration of the surge, which for a single violent event is hours and for a disaster is days, as *Response Forces and Stations* §4.7 states it from the response side.

This category is characterized empirically by the Las Vegas hospital surge of 1 October 2017, conferred in the Emergency Services series as DBA-ESS-RE-11 and cited here by the short title *Route 91 Harvest Shooting* (Section 4.1). The framework records its receiving-side facts from the Nevada Hospital Association's case study and they are not restated here beyond what the requirements above cite (framework §6.4).

5.7 Loss of a Shared Service on Which Care Depends

Postulated event loss of a service the hospital does not run but cannot give care without: a shared pathology and transfusion laboratory, a regional imaging service, a claims and payment clearinghouse, or a supplier of a critical consumable.

Consequence threshold care that stops or slows across every hospital the service supports, at once (Chain C), with the elective care postponed to keep urgent care going.

Required function the hospital must state each shared service its care depends on, the continuity window of the care each supports, and the provision it holds against the service's loss: a second source that does not share the first's systems or supplier, an in-house capability for the most time-critical part, or a manual path. A shared service is credited only where its own design basis covers the event; where it does not, its loss is a design-basis event for every hospital it serves. The service itself is the class that produces it, and the medical-materials class's where it supplies a material (Section 8.3).

Design-basis duration until the service is restored or replaced.

The ransomware attack on Synnovis, the pathology provider to two of London's hospital trusts, on 3 June 2024 took out blood testing and the matching of blood for transfusion across the hospitals it served. By September more than ten thousand outpatient appointments and seventeen hundred elective procedures had been postponed, on the provisional figures of the National Health Service in London, and in June 2025 one of the trusts said the attack had contributed to the death of a patient. The design-basis reading is that the hospitals' continuity rested on a supplier's systems they did not run, and that the event reached every hospital the supplier served together.

5.8 Natural-Hazard Exceedance and Evacuation

Postulated event flood, wind, earthquake, wildfire, extreme heat or cold exceeding the design parameters of the building, its plant or the provision of Sections 5.1 to 5.3, or making the hospital unreachable.

Consequence threshold loss of the building's habitability or its plant (Chain C), and evacuation, which moves the hospital's patients onto its neighbors (Chain S).

Required function the building, its plant and the provision of Sections 5.1 to 5.3 must remain within their design basis under the design-basis natural hazard, with particular attention to a hazard that takes the grid, the water and the roads together. The hospital also states what it does when the hazard exceeds that basis: the point at which it will shelter in place and the point at which it will evacuate, decided in advance with the authority that orders each; the time to move every patient, band by band, to a receiving facility by means that survive the event; and the receiving facilities that have agreed to take them. Federal conditions require arrangements with other facilities to receive patients and policies for evacuation and sheltering in place (42 CFR

482.15(b)). The time to evacuate is part of the continuity window for the patients who cannot be moved quickly, and the decision to evacuate is taken in time for it.

Design-basis duration event-specific; until the building is habitable or every patient has been moved.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Loss of electric power beyond on-site endurance	Outage beyond fuel; generators, fuel pumps, switches or switchgear lost to the event	Chain C	§4.1 candidate (loss of power across the continuity window)
Loss of water supply or pressure	Municipal supply or pressure lost or contaminated; plant dependent on it	Chain C	§4.1 candidate (dialysis-grade water); §6.1
Loss of medical gas or its supply	Piped system failure, bulk loss, resupply interrupted	Chain C	§4.1 candidate (medical gas)
Loss of the hospital's information systems	Record, laboratory, imaging, pharmacy and ordering lost by failure	Chain C + Chain S	§4.1 candidate (cyber compromise); developed here
Cyber attack on clinical and hospital systems	Encryption, destruction or alteration of systems, plant controls or devices	Chain C + Chain S	§4.1 candidate; §5
Mass-casualty surge beyond capacity	More critically injured than the hospital can carry to definitive care; surge without notice	Chain S	§4.1 candidate; §6.4
Loss of a shared service	Shared laboratory, imaging or supplier lost across hospitals	Chain C	developed here
Natural-hazard exceedance and evacuation	Flood, wind, earthquake, fire, heat or cold beyond design; evacuation	Chain C + Chain S	developed here

6. Military-Action Extension: DBA-MA for the Hospital

6.1 MA Extension Applicability Determination

Every hospital in the class performs the facility-level applicability determination the series makes mandatory (framework §7). The primary factor is geolocation relative to military-action

exposure, and the secondary factor is strategic value. The determination is time-varying and re-triggered on the geopolitical indicators the DBA-MA parent framework defines. The framework states a strong sector-level presumption toward applicability for trauma and burn capacity, and names mass-casualty surge as itself a military-action consequence of attacks on other sectors (framework §7). For this class, strategic value is driven by three things: the trauma, burn and specialty capacity the hospital holds that no other in its area can replace; its place as the receiving hospital for a population, a military installation or a site whose attack would produce casualties; and its dependence on a substrate that an attack on another sector would take. The determination is recorded, and any scope reduction carries a recorded basis (Section 6.10).

6.2 The Four Characteristics Applied to the Class

Applied to the hospital, the four military-action characteristics take specific form. Intentionality: the adversary may select the hospital itself, or select a target elsewhere because of the casualties it will produce and the hospitals that must receive them; in either case the hospital's purpose of caring for the injured becomes part of the adversary's calculation. Persistence: the adversary holds access to the hospital's information systems across the interval needed to choose the moment, as the attacker in the Irish health service did for about eight weeks before detonation, on the post-incident review's account, or strikes a region's energy supply repeatedly over a winter so that its hospitals run on generators for weeks. Precision: the adversary chooses the hour, the season and the coincident conditions that carry the harm farthest, an attack on the grid in cold weather, a strike timed to a surge already under way. Denial: the adversary takes the region's power, water and communications before or with the act, so that the hospital receives its casualties with its substrate degraded, its notification cut and its resupply and transfer paths closed. The class validation standard is the state actor who knows what a region's hospitals depend on and can reach it, and who has chosen the day.

6.3 The Surge Under Grid and Communications Denial as the Governing MA Scenario

The governing military-action scenario is a mass-casualty surge that reaches the area's hospitals while the region's grid and communications are denied by the same campaign. It opens both chains at once. The hospitals are on their own power for longer than their essential electrical systems were sized for, with fuel delivery uncertain; the notification of Section 4.4 does not arrive, so each hospital learns of the event from its first patient; the count of Section 4.3 cannot be sent by the paths the hospitals use every day; and the surge arrives by every means, untriaged, while the hospitals' information systems may also be lost. No credit is taken for notification, resupply, transfer or mutual aid that the same campaign can deny. The extension therefore requires three things. It requires that the continuity window for life-sustaining and time-critical care be held through the surge, with the surge's own consumption of power, oxygen, blood and supplies counted against it. It requires that the hospital reach surge posture from its first patient, without notification, and receive at a throughput stated for the staff it would actually have,

recalled by means that do not depend on the networks the campaign denies, which Section 4.6 states as a standing rule of self-activation. And it requires that the hospitals of the area be able to report their counts and throughput to command, and to each other, by a path that works without the grid and the public networks, which Section 4.3 requires each hospital to name.

The grid strikes in Ukraine in the autumn of 2022 show the hospital as the downstream consumer of a grid under deliberate attack. Hospitals ran on generators through repeated outages; surgeons operated by headlamp in Lviv and at the Kyiv Heart Institute; the country's health minister said hospital generators were designed to run for three to four hours against outages that could last up to three days, as reported by the Associated Press. The design-basis reading is that the continuity window a hospital sizes for a storm is not the window a campaign against the grid will ask of it, and that the campaign is counted in weeks. DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*) drives this scenario through the class (Section 9.2).

6.4 The Hospital Struck

The second military-action scenario is a strike on the hospital itself. A hospital is identifiable and fixed, and the protection the law of armed conflict gives it does not prevent the attack; it makes the attack unlawful. The World Health Organization reported in May 2026 that it had verified more than three thousand attacks on health care in Ukraine since February 2022, most of them on facilities. The design basis cannot make a hospital proof against a missile. It can decide what the hospital does in the minutes after one: which patients are moved first and to where, which services the hospital can still give from its undamaged buildings, and how its patients and staff reach shelter when an attack is warned. The class requires a hospital with a positive applicability determination to state its shelter for patients who cannot be moved quickly, the order in which it would move the rest, and the services it would continue from each building if another were lost.

This scenario is characterized empirically by the strike on the Okhmatdyt National Children's Hospital in Kyiv on 8 July 2024, conferred by the framework as DBA-HPH-RE-3 (framework §9.1). The United Nations Human Rights Monitoring Mission in Ukraine assessed a high likelihood that the hospital suffered a direct hit, which it attributed to a Russian missile, during a mass missile attack on the city. About 670 children and more than a thousand staff were reported present; two people were killed, a doctor among them, and dozens wounded; one building was destroyed and others damaged, as reported. The design-basis reading is that a tertiary hospital holding patients who cannot leave was struck in a city under a mass attack, and that its patients had to be moved to other hospitals already working under the same campaign.

6.5 The Coordinated Cyber Scenario

The governing cyber scenario is the Irish health service attack timed to a surge. The hospitals' information systems are encrypted or their records altered as the casualties of an attack elsewhere arrive, so that the hospitals receive on paper, cannot see results, and divert to neighbors whose systems may be lost as well. Against this scenario the class relies on the manual mode of Section 5.4, held ready and measured; on the separation of the plant and its controls from the clinical

networks (Section 5.5); and on the receiving function of Section 4, which is designed to work without the information systems. The scenario has a second form that needs no encryption at all: altered records, orders or device settings that lead clinicians to act on false information while the systems appear normal. That form is governed by the integrity of the basis on which a clinical action is taken, and the obligation is *Indication Integrity*'s (Section 7.3).

6.6 Protected Status and Identification

A hospital is meant to be seen. The Fourth Geneva Convention provides that civilian hospitals may in no circumstances be the object of attack and that they may be marked with the distinctive emblem (Article 18), and Additional Protocol I extends the protection to medical units generally (Article 12). The emblem that tells an attacker to spare the hospital also tells an attacker where it is, as the dam class found of the marking of works containing dangerous forces (*Dam Structures and Water-Control Works* §6.5) and the chemical framework of hazard labels (*Chemical Sector Framework* §7.1). The design-basis response is the same: the identification stays, because the population and the responders need it and because the protection depends on it, and the hospital's design basis assumes the adversary knows where it is. The same Protocol asks the parties, whenever possible, to site medical units so that attacks on military objectives do not imperil them (Article 12(4)), which makes the hospital's location a design-basis matter in its own right. The class does not rely on the law of armed conflict to protect a hospital; it records the law as part of the setting of the military-action extension.

6.7 Cross-Sector Interaction

Under military action the hospital is tied to other sectors in both directions. Coming in, an attacker may take the grid, which takes the hospital's power and, through it, the plant; the water system, which takes dialysis, sterilization and, where the plant depends on it, heat; the public networks, which take notification, the count and staff recall; and the roads, which take fuel delivery, resupply and transfer. Going out, the hospital's failure to receive reaches the response forces, whose units wait at the door, and the population the response serves. A dam failure upstream puts a hospital in its footprint among the places whose occupants cannot easily move (*Dam Structures and Water-Control Works* §4.5). The adversary need not attack the hospital to defeat it; an attack on what it consumes, timed to the casualties of another attack, is enough.

6.8 The Adequacy Result: Autonomous Care Continuity Condition

The test of whether a hospital's protection is adequate is whether it can reach what this document calls the Autonomous Care Continuity Condition. When grid power, municipal water, the public networks, its own information systems and outside resupply are lost, the hospital holds life-sustaining and time-critical care for the continuity window it states, on its own. Its essential electrical system carries the load from protected components and stored fuel; its water, medical gas and plant serve that care from on-site provision; its manual mode carries the clinical functions its information systems carried; its plant and gas alarms work without the clinical

networks; and it receives, at its stated throughput, a surge that arrives without notice, and reports its count by a path that does not depend on the public networks.

The protective functions on which this condition rests are to be locally sovereign as FD-LS defines the property where they are protective functions in that sense (*Local Sovereignty* §3.2): the transfer of the essential electrical system to its generators on the loss of normal power, the automatic changeover of the medical gas supply to its reserve, and the mechanical and electrical limits of the plant. Each is to be shown to be a measurement against a setpoint, colocated with what it protects, and no system an attacker could reach should be able to stop, delay or trigger it, or alter its settings except through a governed act. *Local Sovereignty* governs and is not restated here. Local sovereignty describes what protects the hospital's plant when its control systems have failed. It does not decide whether any clinical action needs human authorization (*Local Sovereignty* §4).

The hospital's staff must also be able to hold this condition for a set period without outside support. That requirement is written here in this document's own words, as the dam class writes its own (*Dam Structures and Water-Control Works* §6.7), because whether the DBA-MA parent framework's local-sovereignty principle is the same thing FD-LS defines is still an open question (Section 10.5). The staff's endurance is the limit that matters most: a hospital working by hand through a surge on generator power exhausts its people before its fuel, and the hold time of the manual mode (Section 5.4) is set by them.

The end state may also be reached by evacuation. Moving the patients ends the hospital's continuity obligation and transfers its demand to the receiving facilities, at a cost the patients least able to be moved bear. Like the drawdown of a dam, it is a protective action with an authority named in advance, a trigger stated in advance, and a statement of what it costs and to whom, and it is taken in time for the slowest patients to be moved (Section 5.8).

6.9 Adequacy and the Declared Residual

In the non-adversarial domain, the Autonomous Care Continuity Condition is achievable with the Section 5 requirements: an essential electrical system whose components survive the event and whose endurance is stated, water, gas and plant provision stated band by band, a manual mode measured in exercises, and a receiving function that works without notice. Against a deliberate attacker, adequacy has limits. Against a cyber attack, it rests on the manual mode and on the separation of the plant from the clinical networks. Against a campaign on the grid, it rests on fuel and on the staff who must work through it. Against a strike, it rests on shelter and evacuation, because a building that must be open to the injured cannot also be hardened against every means of destroying it.

Three residuals are declared, in the output form of *Defense in Depth* §3.5. First, a direct strike on a hospital can destroy any part of it; beyond the design act the residual is care the hospital cannot give, and the response is shelter, evacuation and the capacity of other hospitals, not autonomy. Second, a campaign on the grid or the fuel supply that outlasts the hospital's endurance ends its

continuity; beyond that point the residual is an evacuation the hospital cannot avoid, and the design response is an endurance sized for the campaign's duration rather than a storm's, with the evacuation planned for the moment the endurance runs out. Third, a surge larger than the area's hospitals can carry together leaves critically injured patients waiting; beyond the area's combined throughput the residual is a shift to crisis standards of care, which the federal preparedness guidance describes as care with a significant impact on the usual standard (Assistant Secretary for Preparedness and Response, Technical Resources, Assistance Center, and Information Exchange, surge definitions, 2016), and the design response is the throughput itself and the distribution of patients the two series share. These residuals bound the class adequacy claim.

6.10 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as a continuity window shorter than the most time-critical band requires, an endurance credited on a fuel delivery the event may deny, a throughput credited on staff who would not be present, or a surge assumed to arrive with notice and field triage, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Section 9.1 records it for the framework to adopt, as the transportation framework has.

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

The hospital carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4.

7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the hospital relies on to hold its patients' care independently of its networked systems. These are the essential electrical system and its automatic transfer to the generators, which the codes the federal conditions adopt require to restore power to the life safety and critical branches within ten seconds; the medical gas systems and their alarms; the mechanical and electrical protections of the plant; and the integrity of the indication on which clinicians and engineers act, from a patient monitor to a fuel gauge. An automatic transfer switch that measures the loss of normal power and transfers on its own measurement, with no networked controller able to inhibit or command it, is the clearest candidate for a locally sovereign function in the class. A function meets *Local Sovereignty* §3.2 only if no system an attacker could reach can stop, delay or trigger it, or change its settings

except through a governed act. A building management system that can command the plant from the hospital's business network fails that test for every function it can command.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Controllers and devices that behave predictably within a stated range of inputs, and that have been formally tested as installed, are exempt from the Command Broker requirement for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line: an infusion pump shown to meet that test as installed takes, in delivering a programmed dose, an action whose consequence can kill, and it is outside the obligation only because, and only while, its behavior is bounded by that showing (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act, as the production class states for its safety controllers (*Chemical Production and Synthesis* §7.2). A drug library pushed to a fleet of pumps over the network is such a change. A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence.

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems in the hospital — clinical decision support, diagnostic imaging interpretation, sepsis and deterioration prediction, triage scoring in the emergency department, automated medication and infusion control, bed and diversion management, and generative assistants that draft orders and notes — are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The framework makes the coordination class, OpsCo, the locus at which the sector's Bright Line placement and Command Broker obligation are implemented (framework §3.5), and states that the placement of a decision-support system is made on the consequence of the action it supports, in the series that owns the action (framework §6.4). The actions taken inside the hospital are this class's, and this document names them. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3), and the qualified clinician who authorizes an above-line action is the Command Broker the framework names (framework §5).

Four cases illustrate the application, and none of them is a placement by component. A system within scope that sets or changes an infusion rate takes an action with every change, and each change is placed on its consequence: a change whose consequence can reach the severity threshold for the patient is above the line and requires the authorization of a qualified clinician, and a change bounded by a drug library whose hard limits the hospital's pharmacy and therapeutics authority has approved may be discharged in advance under *Mode of Discharge*, where the library and its limits form a pre-approved envelope and the gatekeeper that enforces them is verified. A triage score that assigns a patient a place in the queue for definitive care during a surge is an action placed on the consequence of the delay it can impose, and where the

score is accepted by a clinician with a single keystroke, the acceptance and the score together are one action. A generative assistant that drafts an order is within scope: the order it proposes, together with the clinician's acceptance of it, is one action placed on the consequence of the order. A system that recommends diversion is placed on the consequence of the diversion (Section 4.5).

Two further points apply across the cases. The federal regulator's guidance on clinical decision support software, revised in January 2026, turns in part on whether the software enables a clinician to review the basis of its recommendation independently, and discusses automation bias (Food and Drug Administration, *Clinical Decision Support Software*, final guidance, January 2026). Whether a system is a regulated device does not decide its placement here, and a system outside the device definition is placed on its consequence like any other. And any output on which an above-line clinical action may be grounded carries the indication-integrity obligation of *Indication Integrity*: a deterioration score computed from a record that may have been altered is not a basis for authorization until the record can be trusted (Section 5.5).

7.4 The Interaction Point: Care Continuity Under Denial

The two design bases meet when the hospital must keep its patients' care going while under attack. The clinical and plant systems must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That obligation reaches each such action whose consequence meets the threshold, and the actions most likely to do so are a change to the therapy a patient in life-sustaining care receives, a change to the settings of a fleet of devices, and a command to the plant that could take power, gas or habitability from the patients in it; each is placed on its own consequence. And the hospital's care must continue when no outside system or authority can be reached at all. The physical design basis guarantees the plant; the manual mode guarantees the clinical functions; the AI-governance design basis guarantees that no proposing system can change a patient's therapy or the hospital's plant without a qualified human authorization the architecture can enforce. When the systems fail, the clinicians act by hand under their own authority, which is where the authority lay all along.

7.5 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, and the Command Broker family under its register. Where the two are cited together, the physical requirement governs the hospital's protective architecture and the AI-governance requirement governs the authorization of any proposing system that interacts with it. The clinical standard a decision-support system must meet is this sector's to set (framework §6.4) and is not set here; its development belongs with the coordination class.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On: the Receiving Hospital

Two sets of people depend on the hospital. The first are the patients inside it, whose care Chain C measures. The second are the people still on their way to it: the injured of every incident in its area, the patients the response forces carry, and the patients its neighbors divert to it. The Emergency Services series depends on the hospital at the patient-transfer point, and its response forces cannot clear a scene faster than the hospitals can receive what they bring (*Response Forces and Stations* §4.7). Patients who depend on dialysis, oxygen or other care at home depend on the hospital when their own supply fails, and in a regional outage they arrive in numbers. The class requires the hospital's design basis to count them in its surge.

8.2 What the Hospital Depends On

The hospital depends on electric power, water, medical gas and fuel, communications, its information systems and their suppliers, the supply of medical materials, and the roads (framework §6.1). Electric power is the dependency the electric series already elevates, carrying hospitals at its Tier-3 life-safety designation (framework §1.5), and its loss beyond the essential electrical system's endurance is the design-basis condition of Section 5.1. Water is the dependency the water series already elevates for hospitals and dialysis (Water Sector Framework §8.2), and the framework states the elevation as reciprocal: a water, electric or communications asset that is the sole supply path for a hospital inherits the hospital's life-safety tier (framework §6.3). The class gives each such asset the continuity window of Section 3.3, so that its own design basis can count it. Communications carry notification, the count, staff recall and the hospital's links to its suppliers. Medical materials and blood come from the medical-materials class. Fuel and resupply come by road and are the transportation series'. A hospital may also depend on a dam upstream, as a facility in its footprint (*Dam Structures and Water-Control Works* §4.5, §8.1).

8.3 The Boundaries with the Partner Classes and Series

With the Emergency Services series, the framework draws the line at the patient-transfer point and divides medical control on the same line (framework §6.4); this class states the receiving side at Section 4, and the response side is *Response Forces and Stations* §4.7, §6.5 and §8.3 and *Public-Safety Answering and Dispatch* §8.3. The ambulance and its crew are the Emergency Services series' until the patient is handed over; the emergency department is this class's from that moment. With *Medical Materials and Supply*, the production and supply of a material is that class's and the hospital's stock and its continuity window are this class's. With *Public Health and Laboratory*, a public-health laboratory is that class's, and a hospital's own laboratory is this class's; a shared clinical laboratory service run for hospitals by a third party is offered to the framework for placement (Section 10.6). With the coordination class, the hospital's own incident command is this class's, and the coordination of hospitals across a region, through health-care

coalitions, the federal preparedness structure and the health information exchanges, is OpsCo's (framework §3.5).

8.4 Regulatory and Institutional Context

The class sits under an established framework of conditions and accreditation. Hospitals that take part in Medicare and Medicaid must meet the federal emergency preparedness condition, issued in September 2016 and in force for hospitals from November 2017 (81 FR 63860, 16 September 2016; 42 CFR 482.15). It requires an all-hazards emergency plan based on a facility-based and community-based risk assessment, policies and procedures, a communication plan, and training and testing, each reviewed at least every two years since the burden-reduction rule of 2019 (84 FR 51732, 30 September 2019); two exercises a year, one of them a full-scale or functional exercise; and the emergency and standby power provisions of paragraph (e). It incorporates the 2012 editions of the health-care facilities code and the life safety code and the 2010 edition of the standard for emergency and standby power systems (42 CFR 482.15(h)). The Joint Commission's emergency management standards, revised for hospitals from 1 July 2022, organize the plan around communications, resources and assets, safety and security, staffing, utilities and patient care, and require the ninety-six-hour plan of Section 5.1. Dialysis facilities meet their own conditions (42 CFR 494.40, 494.62). The duty to screen and the rules on diversion and transfer are the Emergency Medical Treatment and Labor Act's (42 USC 1395dd; 42 CFR 489.24). Trauma centers are verified against the American College of Surgeons' standards. The federal Hospital Preparedness Program funds health-care coalitions and measures their surge through its exercises.

Security is less settled than safety. The emergency preparedness condition requires the hospital to plan for the hazards its risk assessment identifies and does not set a cyber requirement of its own. The HIPAA Security Rule requires a contingency plan for electronic health information, including an emergency mode operation plan (45 CFR 164.308(a)(7)(ii)(C)), which is the nearest federal requirement to the manual mode of Section 5.4, but it sets neither a measured manual rate nor the separation of the plant from the clinical networks. The class records that the cyber requirements of Sections 5.4 and 5.5, the measured manual mode and the separation of the plant from the clinical networks, are not required by any federal condition this instrument has found for hospitals, and that their status is to be confirmed at each revision (Section 10.8).

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments inherit the framework's methodology and architecture and add their facility-specific consequence chain and boundaries (framework §8.1); Table 9-1 states that inheritance element

by element. Where the framework is a skeleton, the class content is developed ahead of it and is marked for the framework to take up.

Table 9-1. Inheritance map for DBA-HPH-CD-FC1.

Element	Source in the framework	Inheritance mode
Function-primary placement, CD function class	§1.3, §3.1, §3.2	Inherited unchanged
Time-criticality sub-index	§3.1, §8.2	Inherited with adaptation; three bands stated, dialysis and long-term care at sub-index depth (Section 2.2)
Design-basis principles (consequence severity, not likelihood)	§4.1	Inherited unchanged
Patient-care continuity window	§3.6 (proposed)	Calibrated here by band and substrate (Section 3.3); for the framework to take up
Surge throughput and time to surge posture	§6.4 (evidence)	Locally defined on the author's decision of 12 October 2026
Event categories	§4.1 (candidates)	Developed here ahead of the framework; for the framework to take up
Bright Line and Command Broker architecture	§5; FD-BL; Command Broker family	Inherited with adaptation: actions inside the hospital placed here, by consequence (Section 7.3)
Patient interface with Emergency Services	§6.4	Inherited and developed (Section 4)
Inward dependencies and reciprocal elevation	§6.1, §6.3	Inherited with adaptation; continuity window given to sole-supply assets (Section 8.2)
MA applicability and its presumption	§7	Inherited with adaptation to facility level
Identification as a targeting aid	not stated in the framework	Adapted from the dam class and the chemical framework; for the framework to take up
Local sovereignty of protective functions	FD-LS §3.2 (not the framework)	Cited directly

Element	Source in the framework	Inheritance mode
Independent protection and declared residual	DBA-MA-DiD §3.5 (not the framework)	Cited directly; for the framework to take up
Consequence-basis conditionality check	not stated in the framework	Locally stated; for the framework to adopt
Consequence chains, receiving function and the Autonomous Care Continuity Condition	—	Locally defined

9.2 Anticipated Derivatives

Hospital-, system- and state-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. A regional design basis, written for the hospitals of one area together so that their combined throughput and their mutual receiving arrangements are stated as one, is anticipated beneath this class, and its designator is the framework's to confer. The three reference events this class proposed at v0.1, on the author's decision of 12 October 2026 that the class propose events of its own after research, are conferred by the framework at §9.1 on the author's decision of the same date: DBA-HPH-RE-1, the ransomware attack on the Irish Health Service Executive, May 2021 (Section 5.5); DBA-HPH-RE-2, the loss of power at Memorial Medical Center during Hurricane Katrina, August and September 2005 (Section 5.1); and DBA-HPH-RE-3, the strike on the Okhmatdyt National Children's Hospital, July 2024 (Section 6.4). Each will read its event back through this class's design basis, and their instruments are Project 11's to write. The Las Vegas hospital surge is cited from the Emergency Services series and is not proposed again (Section 4.1). The facility-class use case for the governing military-action scenario of Section 6.3 has issued as DBA-HPH-UC-1 (*A Mass-Casualty Surge During a Grid and Communications Outage*). Of its six findings, two validate the class, three found gaps that this revision takes in at Sections 3.3, 4.3, 4.4, 4.6 and 5.1, and one is noted to the Emergency Services series (Section 10.10).

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The continuity window, surge throughput and time to surge posture of Section 3.3 are stated as parameters, not numbers. Their values await validation with hospitals against the plans, exercises and records they already hold (Section 3.3). Where those plans assume notification, field triage, a grid outage of a storm's length or resupply within days, validation against them does not validate the military-action case, which Section 6.3 sets.

10.2 Design-Basis Durations

The framework's design basis is a skeleton (framework §4) and sets no durations. The durations in Section 5 are stated as class parameters, and the class durations for holding the Autonomous Care Continuity Condition await the framework's values.

10.3 Dialysis Clusters and Long-Term Care

On the author's decision of 12 October 2026 the class is written for hospitals in full, and dialysis clusters and long-term care are stated at the depth of the sub-index (Section 2.2). Their development, each with its own continuity window, reference facility and design-basis events, is for a later revision of this class. Whether either should instead become a class of its own is the framework's to decide, and the register is unchanged until it does.

10.4 The Las Vegas Surge and the Corpus Register

On the author's decision of 12 October 2026, the Las Vegas hospital surge keeps its single designator in the Emergency Services series, DBA-ESS-RE-11, and healthcare is to be entered as an impacted sector in the corpus register. The register is Project 0's, and the entry is noted to it, not made here.

10.5 Local Sovereignty in the DBA-MA Sense

The protective-function property at Section 6.8 cites *Local Sovereignty* §3.2 directly. The endurance requirement beside it, holding care without outside power, communications or support, is the property the DBA-MA parent framework's local-sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is stated locally until then. The item is registered under Project 9 OI-16.

10.6 The Boundary with the Coordination Class and the Shared Laboratory

The framework places the sector's coordination, and the application of the AI-governance architecture across the sector, with OpsCo (framework §3.5), and states that the placement of a decision-support system is made in the series that owns the action (framework §6.4). This instrument places the actions taken inside the hospital here and leaves the sector-wide clinical standard to the coordination class (Section 7.5), and offers that reading to the framework to confirm. It also offers for placement the shared clinical laboratory run for hospitals by a third party, which is neither a hospital's own laboratory nor a public-health laboratory (Section 8.3). Both are listed as open until the framework decides them.

10.7 Reference Events and the Use Case

The three reference events this class proposed at v0.1 are conferred by the framework at §9.1 as DBA-HPH-RE-1 to DBA-HPH-RE-3, on the author's decision of 12 October 2026, and their instruments are owed by Project 11, which owns every reference-event instrument of the corpus. On the author's decision of the same date, Memorial Medical Center is conferred as an event

distinct from the failure of coordination in Hurricane Katrina, which the Emergency Services series holds as DBA-ESS-RE-8: Memorial is the loss of one hospital's power and its consequence to the patients inside, a different consequence chain in the same storm, and its instrument cites DBA-ESS-RE-8 for the regional context rather than restating it. The Paris attacks of 13 November 2015, in which the city's public hospital system activated its white plan and sixteen of its forty hospitals received the injured (Hirsch and colleagues, *The Lancet*, 2015), are held by the framework as a candidate for the surge chain and are not conferred. The use case for the governing scenario of Section 6.3 has issued as DBA-HPH-UC-1, and the three gaps it found are taken in at this revision (Section 9.2). The register rows and the impacted sectors the instruments will propose are Project 0's to enter.

10.8 The Cyber Requirement for Hospitals

The federal emergency preparedness condition sets no cyber requirement of its own (Section 8.4). This instrument has found none in the conditions of participation; the HIPAA Security Rule requires an emergency mode operation plan for electronic health information (45 CFR 164.308(a)(7)(ii)(C)) but sets neither a measured manual rate nor plant separation (Section 8.4). It has found no requirement for the measured manual mode or the separation of the plant from the clinical networks that Sections 5.4 and 5.5 require. The gap is recorded, and its status is to be confirmed at each revision of this instrument.

10.9 The Dialysis Water Standard

The federal conditions for dialysis facilities incorporate an industry standard of 2004 for water and dialysate quality (42 CFR 494.40), while the current consensus standards for dialysis fluids are the ISO 23500 series, whose general requirements were revised in 2024. The gap between the incorporated standard and the current one is recorded for the development of the dialysis sub-index (Section 10.3).

10.10 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the calibration of the continuity window (Section 3.3), the event categories of Section 5, the refinement of the sub-index into three bands (Section 2.2), the identification finding of Section 6.6, the declared residuals of Section 6.9, the conditionality principle of Section 6.10, and the placement reading of Section 10.6; the conferral section it also asked for is now framework §9.1. To the Emergency Services series: that the receiving side of the patient interface is stated at Section 4 to match *Response Forces and Stations* §4.7, and that a hospital's notification path should record receipt (Section 4.4), for *Public-Safety Answering and Dispatch* to read beside its own; and, from DBA-HPH-UC-1, that a hospital whose path to command under denial is a terminal on the area's public-safety radio is a user of that system, for the answering class to decide whether to count the receiving hospitals among the users its radio serves under denial (Section 4.3). To the water

series: that a hospital's plant can depend on municipal pressure for its heat and cooling, so that the elevation the water framework gives hospitals reaches the plant room as well as the clinical areas (Section 5.2). To the electric series, in Project 8: that the continuity window of Section 3.3 is the figure its life-safety tier should be measured against, and that the components between a generator and its load are part of the hospital's endurance (Section 5.1). To the Dams series: that a hospital in a dam's footprint is among the places whose occupants cannot easily move, which *Dam Structures and Water-Control Works* §4.5 already counts. And to Project 0: the impacted-sector entry of Section 10.4.

References

DBA-HPH Series and Parent Framework

Healthcare and Public Health Sector Framework (DBA-HPH_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics and the design-basis-under-military-action methodology.

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

Response Forces and Stations (DBA-ESS-RF-FC2) and *Public-Safety Answering and Dispatch* (DBA-ESS-AD-FC1). Eclectic Technologies, 2026. The response side of the patient interface and of medical control.

The Route 91 Harvest Shooting of October 2017 (DBA-ESS-RE-11). Eclectic Technologies, 2026. Held by Project 11; cited by the short title *Route 91 Harvest Shooting* and finding number.

The Failure of Coordination in Hurricane Katrina (DBA-ESS-RE-8). Eclectic Technologies, 2026. Held by Project 11; cited for the regional context of DBA-HPH-RE-2.

DBA-HPH-RE-1, DBA-HPH-RE-2 and DBA-HPH-RE-3, conferred at framework §9.1 on 12 October 2026. Their instruments are owed by Project 11.

A Mass-Casualty Surge During a Grid and Communications Outage (DBA-HPH-UC-1). Eclectic Technologies, 2026. The use case for the governing military-action scenario of Section 6.3, whose findings this revision takes in.

Water Sector Framework (DBA-W-Sector_Framework). Eclectic Technologies, 2026. The dialysis dependency (§8.2).

Dam Structures and Water-Control Works (DBA-DAMS-DS-FC1), *Chemical Production and Synthesis* (DBA-CHEM-PR-FC1), *Chemical Sector Framework* (DBA-CHEM_Sector_Framework) and *Nodes and Terminals* (DBA-TRAN-ND-FC1). Eclectic Technologies, 2026. The structural model for this instrument and the precedents it follows for identification, the determinism posture and the manual fallback rate.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Guidance

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Geneva Convention relative to the Protection of Civilian Persons in Time of War (Fourth Geneva Convention), 12 August 1949, Article 18. Protocol Additional to the Geneva Conventions of 12 August 1949, and relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977, Article 12.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-HPH series, typed physical under the author’s determination of 30 September 2026, in the ten-section structure of DBA-DAMS-DS-FC1 and DBA-CHEM-PR-FC1. Written on the author’s four decisions of 12 October 2026, and checked independently before issue against primary sources and the framework, every item taken: the class is DBA-HPH-CD-FC1 as the framework’s register states it, with hospitals as receiving facilities in full and dialysis clusters and long-term care at the depth of the sub-index; the two chains, continuity and surge, measured by the patient-care continuity window and by surge throughput to definitive care, with the time to surge posture as a further parameter; the Las Vegas surge cited from DBA-ESS-RE-11 with healthcare as an impacted sector rather than conferred again; and events of the class’s own proposed after research. Establishes three reference facilities (a Level I trauma center, the community hospital nearest a likely scene, and a multi-hospital system on one shared record); the receiving function, developing the hospitals’ side of the patient interface the framework states at §6.4; the non-adversarial event envelope; the military-action extension with the surge under grid and communications denial as its governing scenario; the Autonomous Care Continuity Condition and its three declared residuals; the dual design basis stated in full; and the required inheritance map.

Version	Date	Author	Description
			Proposes three reference events for the framework to confer: the Health Service Executive ransomware attack 2021, Memorial Medical Center in Hurricane Katrina 2005, and the Okhmatdyt strike 2024, each drawn from the primary or best available sources that could be reached, and records the Katrina overlap and a Paris candidate for the author. Quantitative parameters and durations are flagged as pending.
0.2 — DRAFT	October 2026	Tim Roxey	Conforms to the framework at v0.6 and takes in the three gaps DBA-HPH-UC-1 returned. The three reference events proposed at v0.1 are stated as conferred, as DBA-HPH-RE-1 (Health Service Executive, 2021), DBA-HPH-RE-2 (Memorial Medical Center in Katrina, 2005) and DBA-HPH-RE-3 (Okhmatdyt, 2024), at Sections 5.1, 5.5, 6.4, 9.2 and 10.7, with Memorial distinct from DBA-ESS-RE-8 and Paris 2015 held, on the author's decisions of 12 October 2026. From the use case: Section 3.3 states the surge throughput on generator power, with the loads it needs on the branches that carry them, and Section 5.1 states endurance at that load; a new Section 4.6 states a standing rule of self-activation with a labor pool and a designated reserve, to which Sections 3.3 and 6.3 point; Sections 4.3 and 4.4 require the hospital to name the path to command that remains without the public networks. Sections 2.4, 6.3 and 9.2 cite the issued use case. Section 10.10 records the conferral section as made and notes the use case's finding on the area's radio to the Emergency Services series. References add DBA-ESS-RE-8, the conferred designators and the use case. The cover lineage stays as issued. No change to the class's scope, chains, event categories, placements or residuals; the v0.1 entry is left as a record.